Customer Retention Enhancement through Predictive Analytics

Data Gathering, Exploratory Analysis & Data Preparation Report

SmartBank – A Subsidiary of Lloyds Banking Group

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1. Introduction

This report presents the findings from the initial phase of the Customer Retention Enhancement initiative. The objective of this phase was to gather relevant customer data, conduct exploratory data analysis (EDA), and prepare a clean, integrated dataset for predictive modelling.

The ultimate goal of this project is to identify customers at risk of churn and support targeted retention strategies to protect SmartBank's long-term profitability and competitive position.

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2. Data Collection and Integration

2.1 Data Sources

The following datasets were integrated using CustomerID as the primary key:

1. Customer Demographics
    - Age
    - Gender
    - Marital Status
    - Income Level
2. Transaction History
    - Transaction count
    - Total amount spent
    - Average transaction value
    - Product category usage
3. Customer Service Interactions
    - Total service interactions
    - Complaint frequency
    - Resolution status (resolved vs unresolved)
4. Online Activity
    - Login frequency
    - Last login date
    - Service usage

5. Churn Status
    - Binary churn indicator (0 = retained, 1 = churned)

2.2 Data Preparation

To ensure modelling readiness:

- Transaction-level data was aggregated to customer-level summaries.
- Service interactions were summarised to include total interactions, complaints, and unresolved cases.
- Missing service data was interpreted as zero interactions and handled accordingly.
- Last login date was transformed into a behavioural metric: Days Since Last Login.
- All datasets were merged into a single customer-level analytical table.
- Final validation confirmed no remaining missing values.

The dataset consists of 1,000 customers with a churn rate of 20.4%.

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3. Exploratory Data Analysis (EDA)

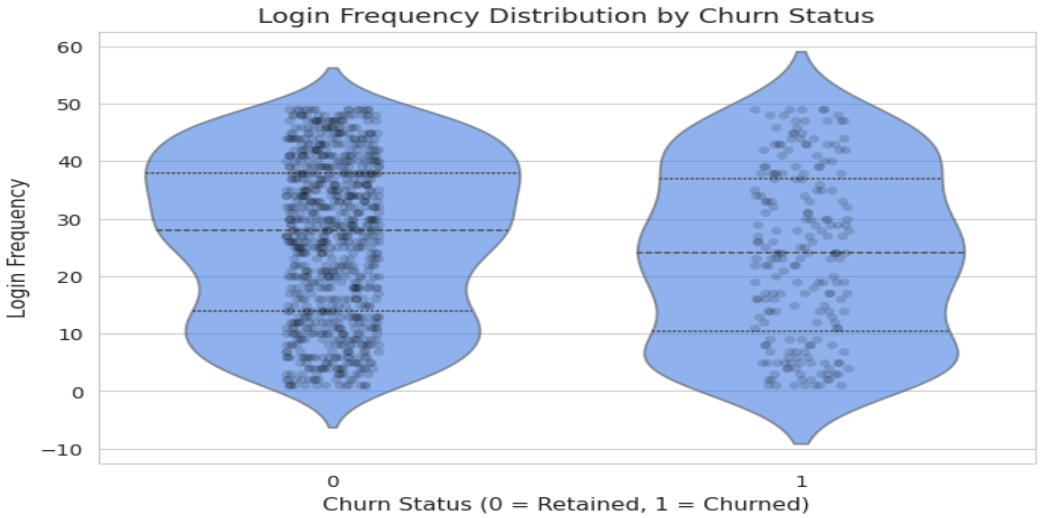
3.1 Churn Distribution

Churn represents 20.4% of the customer base. This reflects a realistic retention scenario and indicates moderate class imbalance.

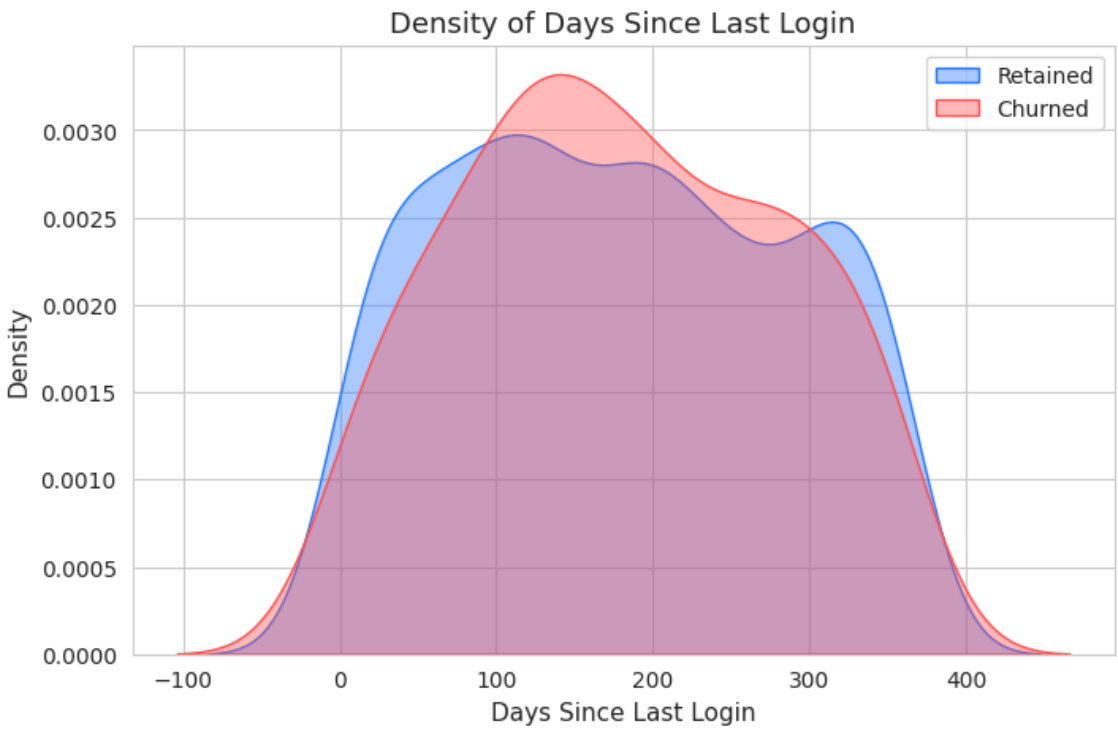
3.2 Behavioural Insights

Customer engagement metrics showed the strongest association with churn:

- Login frequency demonstrated the strongest inverse correlation with churn.
- Customers with lower digital engagement were slightly more likely to churn.
- Increased days since last login suggests behavioural disengagement.



[Figure 1: Login Frequency Distribution by Churn Status]

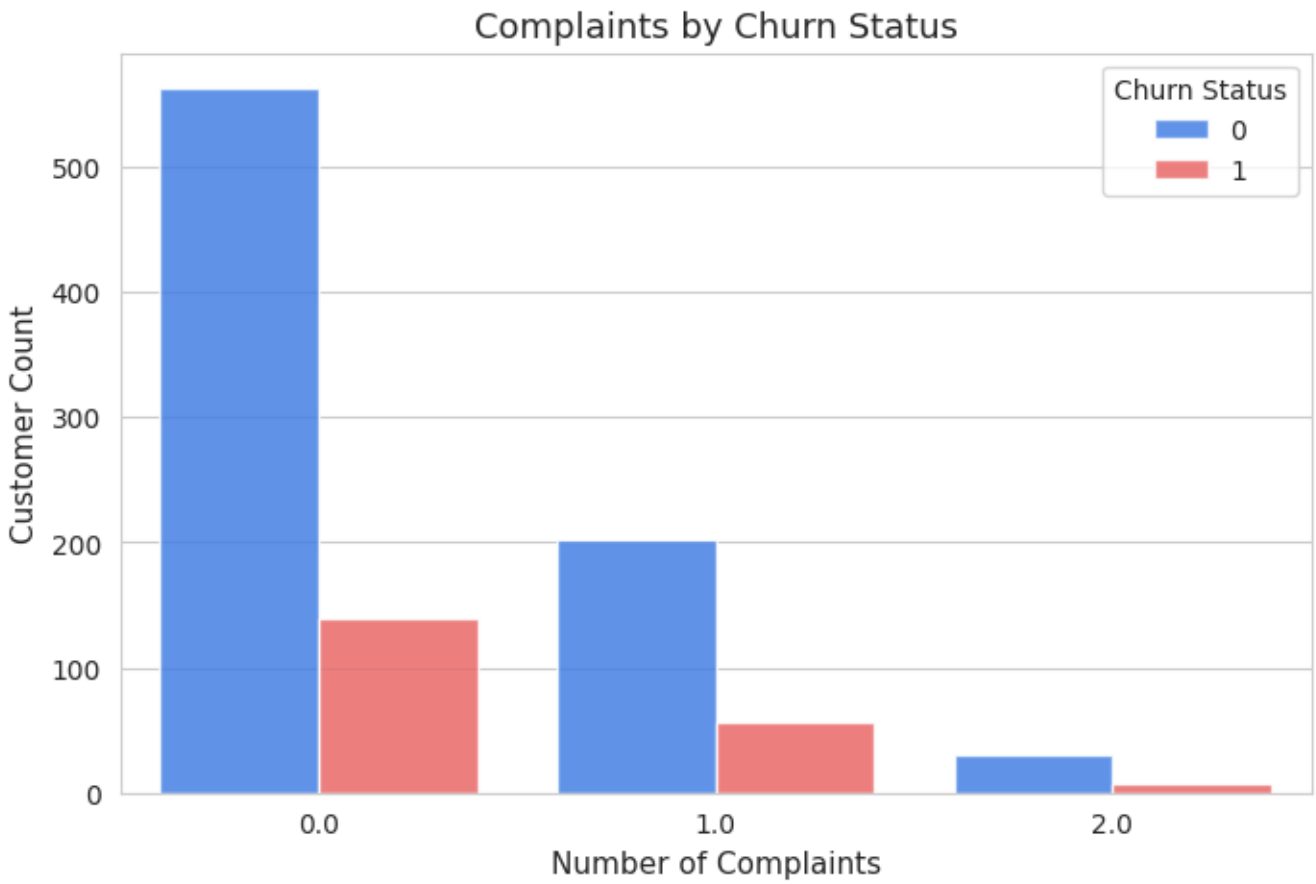


[Figure 2: Density Plot of Days Since Last Login by Churn Status]

This indicates that digital inactivity may serve as an early churn signal.

3.3 Service Interaction Analysis

Complaint frequency and unresolved cases showed minimal standalone separation between churned and retained customers.

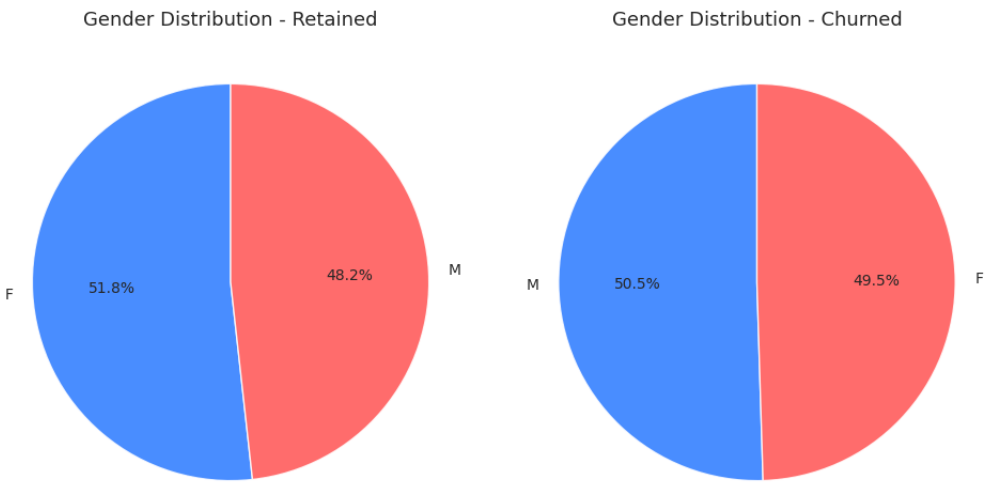


[Figure 3: Complaints by Churn Status]

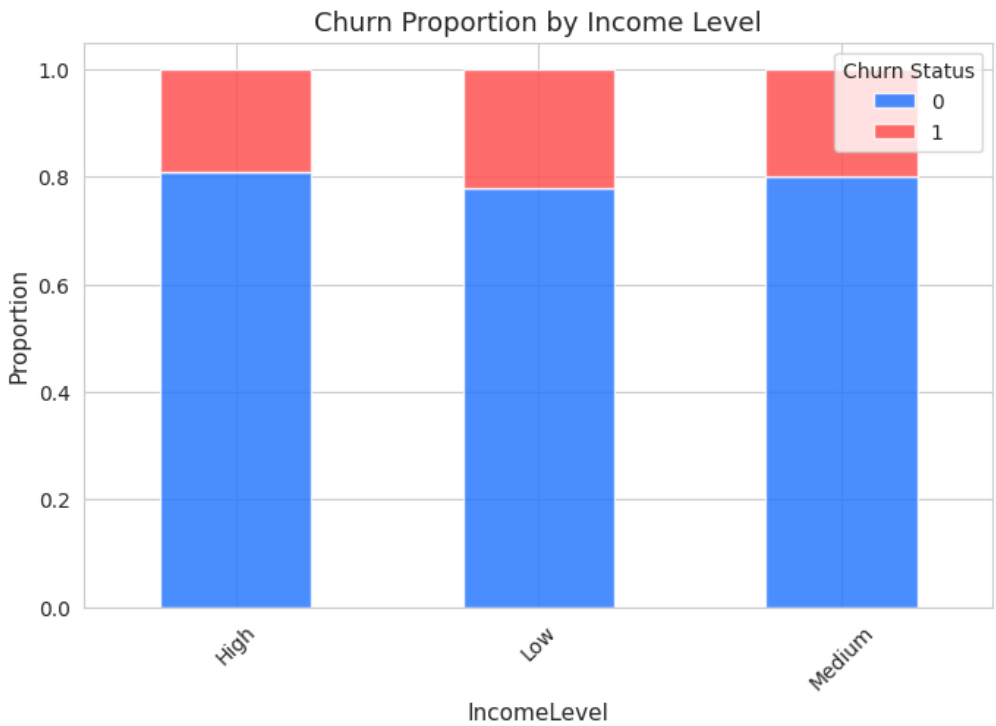
While service dissatisfaction may contribute to attrition, it does not appear to be a dominant independent driver within this dataset. It is likely that service factors interact with behavioural patterns rather than acting alone.

3.4 Demographic Factors

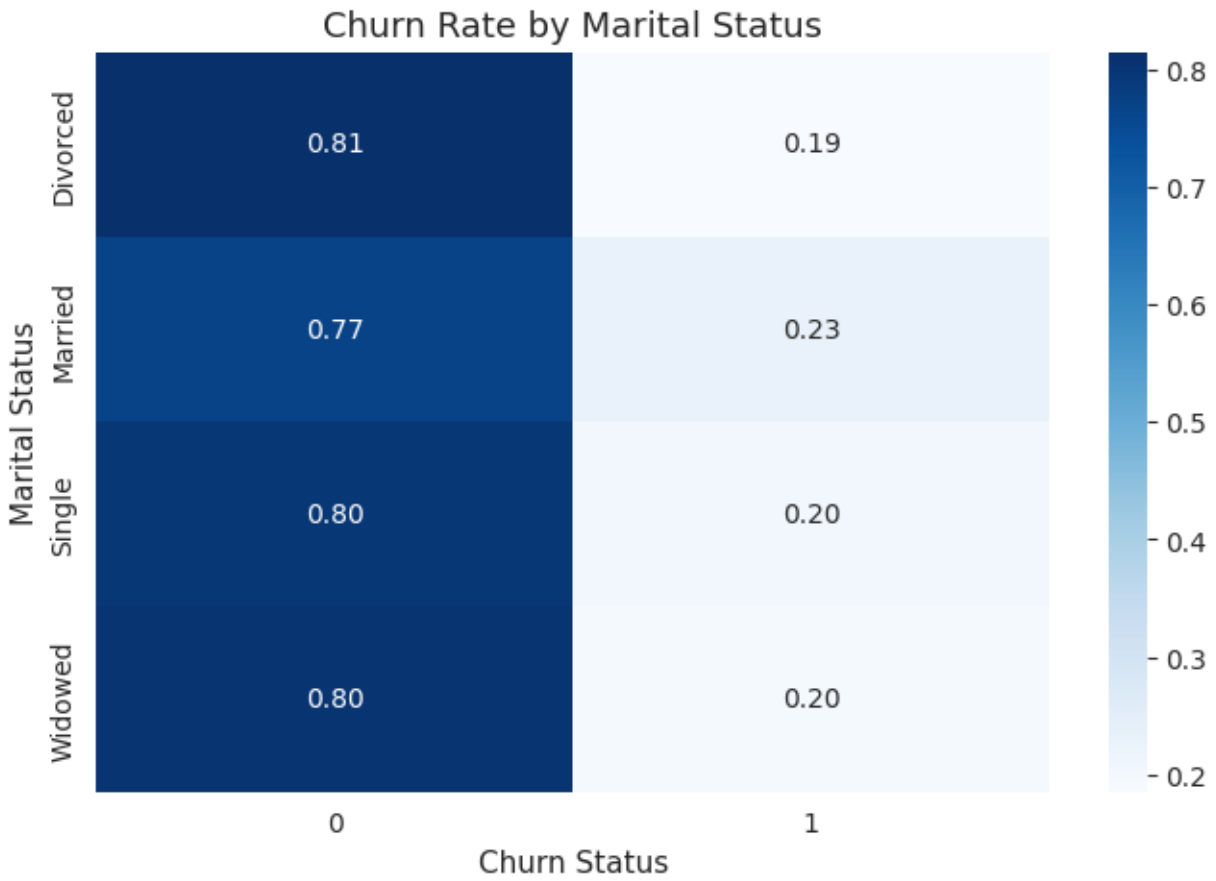
Demographic variables such as gender, marital status, and income level showed limited differentiation between churned and retained groups.



[Figure 4: Gender Distribution – Retained vs Churned]



[Figure 5: Churn Proportion by Income Level]



[Figure 6: Churn Rate by Marital Status Heatmap]

This suggests that churn risk is primarily behaviour-driven rather than demographic-driven.

3.5 Spending and Transaction Behaviour

Total spending and transaction counts displayed subtle differences across churn groups but did not show strong linear relationships.

[Insert Figure 9: Transaction Metrics by Churn Status]

This indicates that financial activity alone is insufficient to predict churn without incorporating engagement metrics.

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4. Key Findings

1. Customer engagement is the most meaningful early indicator of churn risk.
2. Churn behaviour appears subtle and likely influenced by multiple interacting factors.
3. Demographics alone are not strong predictors.
4. A multivariate predictive model is necessary to capture complex patterns.

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5. Conclusion and Next Steps

The exploratory analysis confirms that churn within SmartBank's customer base is not driven by a single dominant factor. Instead, behavioural disengagement appears to be the strongest early indicator.

The cleaned and engineered dataset is now suitable for predictive modelling.

The next phase will involve:

- Developing a classification model to estimate churn probability.
- Evaluating performance using appropriate metrics such as Recall and ROC-AUC.
- Identifying high-risk customers to support targeted retention strategies.